

SEQUENCES & SERIES (NSSCAS)

TEXT BOOK:

$y = mx + c$ (AS): Chapter 2 (p. 194-240)

THE SYLLABUS REQUIRES THE FOLLOWING FROM CANDIDATES:

GENERAL OBJECTIVES <i>Learners will:</i>	SPECIFIC OBJECTIVES <i>Learners should be able to:</i>
Topic 2: Sequences and series	
gain further understanding of sequences and series	- use the expansion of $(a + b)^n$, where n is a positive integer (knowledge of the greatest term and properties of the coefficients are not required, but the notations $\binom{n}{r}$ and $n!$ should be known) - recognise arithmetic and geometric progressions - use the formulae for the n^{th} term and for the sum of the first n terms to solve problems involving arithmetic or geometric progressions - interpret and apply the Sigma notation - use the condition for the convergence of a geometric progression, and the formula for the sum to infinity of a convergent geometric progression

1. FORMULAE

AP's

a : 1st term $\rightarrow T_1$

d : common difference

$$d = T_2 - T_1$$

$$T_2 - T_1 = T_3 - T_2$$

n^{th} term $\rightarrow$ general term T_n

$$T_n = a + (n - 1)d$$

Sum of AP (Series)

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$S_n = \frac{n}{2} [a + l]$$

GP's

a : 1st term $\rightarrow T_1$

r : common ratio

$$r = \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = \frac{T_3}{T_2}$$

n^{th} term $\rightarrow$ general term T_n

$$T_n = ar^{n-1}$$

Sum of GP (Series)

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{for } r < 1$$

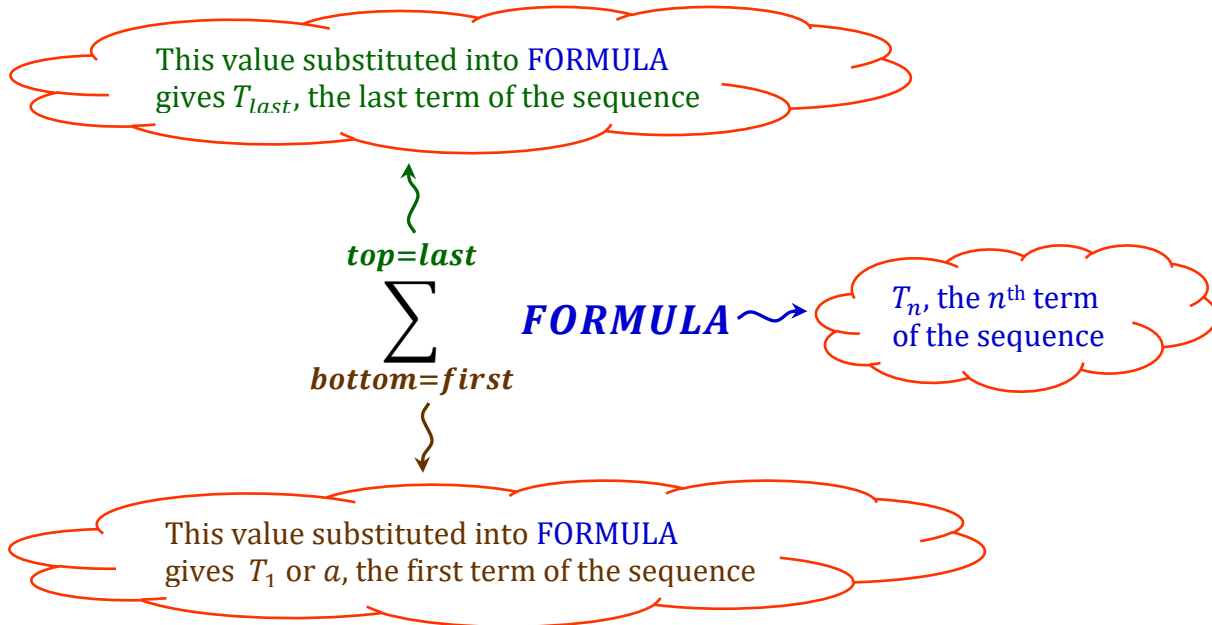
$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{for } r > 1$$

Sum to Infinity GP $\rightarrow$

$$S_n = \frac{a}{1-r}$$

2. SIGMA NOTATION

Σ : “the sum of”



ORDER OF OPERATIONS:

→ Calculate the 1st 3 terms:

- * a or T_1 : Subst. “**bottom value**” into **formula**
- * T_2 : Subst. “**bottom value + 1**” into **formula**
- * T_3 : Subst. “**bottom value + 2**” into **formula**

→ Determine whether sequence is AP or GP:

- * If $T_3 - T_2 = T_2 - T_1 \therefore$ common difference $\sim d \therefore$ AP
- * If $\frac{T_3}{T_2} = \frac{T_2}{T_1} \therefore$ constant ratio $\sim r \therefore$ GP

→ Determine n , the number of elements in the sequence:

- * $n = (\text{top value} - \text{bottom value}) + 1$

→ If AP, use:

$$* S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} [a + l]$$

→ If GP, use:

$$* S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

→ If $n = \infty$, use:

$$* S_n = \frac{a}{1 - r}$$

2.1 FINDING SUM OF SEQUENCE:

AP's

Eg. Evaluate:

$$\sum_{p=6}^{16} (-3p - 4)$$

$$\begin{aligned} \rightarrow T_1 &= -3(6) - 4 = -22 & \therefore a &= -22 \\ T_2 &= -3(7) - 4 = -25 \\ T_3 &= -3(8) - 4 = -28 \end{aligned}$$

$$\begin{aligned} \rightarrow T_2 - T_1 &= -25 - (-22) = -3 \\ T_3 - T_2 &= -28 - (-25) = -3 & \therefore d &= -3 \\ \text{Common difference} &\therefore \text{AP} \end{aligned}$$

$$\rightarrow n = (16 - 6) + 1 = 11$$

$$\rightarrow S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$\begin{aligned} \therefore S_{11} &= \frac{11}{2} [2(-22) + (11 - 1)(-3)] \\ &= 5,5 [-74] \\ &= -407 \end{aligned}$$

GP's

Eg. Evaluate:

$$\sum_{t=3}^{20} \left(\frac{1}{2}\right)^{t-1}$$

$$\rightarrow T_1 = \left(\frac{1}{2}\right)^{3-1} = \frac{1}{4} \quad \therefore a = \frac{1}{4}$$

$$T_2 = \left(\frac{1}{2}\right)^{4-1} = \frac{1}{8}$$

$$T_3 = \left(\frac{1}{2}\right)^{5-1} = \frac{1}{16}$$

$$\rightarrow \frac{T_3}{T_2} = \frac{1}{16} \div \frac{1}{8} = \frac{1}{2}$$

$$\frac{T_2}{T_1} = \frac{1}{8} \div \frac{1}{4} = \frac{1}{2} \quad \therefore r = \frac{1}{2}$$

Constant ratio $\therefore$ GP

$$\rightarrow n = (20 - 3) + 1 = 18$$

$$\rightarrow S_n = \frac{a(1-r^n)}{1-r}$$

$$\begin{aligned} \therefore S_{18} &= \frac{\frac{1}{4} \left(1 - \left(\frac{1}{2}\right)^{18}\right)}{1 - \frac{1}{2}} \\ &= 0,4999 \\ &\approx 0,5 \end{aligned}$$

2.2 FINDING n^{th} TERM:

AP's

Eg. Find the value of n :

$$\sum_{a=1}^n (2a - 3) = 48$$

$$\begin{aligned} \rightarrow T_1 &= 2(1) - 3 = -1 & \therefore a = -1 \\ T_2 &= 2(2) - 3 = 1 \\ T_3 &= 2(3) - 3 = 3 \end{aligned}$$

$$\begin{aligned} \rightarrow T_2 - T_1 &= 1 - (-1) = 2 \\ T_3 - T_2 &= 3 - 1 = 2 & \therefore d = 2 & \therefore \text{AP} \end{aligned}$$

$$\rightarrow S_n = \frac{n}{2} [2a + (n-1)d] = 48$$

$$\therefore \frac{n}{2} [2(-1) + (n-1)(2)] = 48$$

$$\therefore \frac{n}{2} [-2 + 2n - 2] = 48$$

$$\therefore n [2n - 4] = 96$$

$$\therefore 2n^2 - 4n - 96 = 0$$

$$\therefore (2n + 12)(n - 8) = 0$$

$$\therefore n = -6 \quad \text{or} \quad n = 8$$

$$\therefore \neq \text{possible solution} \quad \therefore n = 8$$

GP's

Eg. Find the value of n :

$$\sum_{p=1}^n 4 \left(\frac{1}{2}\right)^{p-1} = 7 \frac{63}{64}$$

$$\rightarrow T_1 = 4 \left(\frac{1}{2}\right)^{1-1} = 4 \left(\frac{1}{2}\right)^0 = 4 \quad \therefore a = 4$$

$$T_2 = 4 \left(\frac{1}{2}\right)^{2-1} = 4 \left(\frac{1}{2}\right)^1 = 2$$

$$T_3 = 4 \left(\frac{1}{2}\right)^{3-1} = 4 \left(\frac{1}{2}\right)^2 = 1$$

$$\rightarrow \frac{T_3}{T_2} = \frac{1}{2} = \frac{1}{2}$$

$$\frac{T_2}{T_1} = \frac{2}{4} = \frac{1}{2} \quad \therefore r = \frac{1}{2} \quad \therefore \text{GP}$$

$$\rightarrow S_n = \frac{a(1-r^n)}{1-r} = 7 \frac{63}{64}$$

$$\therefore \frac{4 \left(1 - \frac{1}{2}^n\right)}{1 - \frac{1}{2}} = \frac{511}{64}$$

$$\therefore \frac{4 \left(1 - \frac{1}{2}^n\right)}{\frac{1}{2}} = \frac{511}{64}$$

$$\therefore 8 \left(1 - \frac{1}{2}^n\right) = \frac{511}{64}$$

$$\therefore \left(1 - \frac{1}{2}^n\right) = \frac{511}{64} \times \frac{1}{8}$$

$$\therefore -\frac{1}{2}^n = \frac{511}{512} - 1$$

$$\therefore -\frac{1}{2}^n = -\frac{1}{512}$$

$$\therefore \frac{1}{2}^n = \frac{1}{2^9}$$

$$\therefore 2^{-n} = 2^{-9}$$

$$\therefore -n = -9$$

$$\therefore n = 9$$

2.3 ADVANCED EXAMPLES

NOTE: n is NOT the n^{th} term $\rightarrow n$ is a variable in the formula of the series

AP's

Eg. Find the value of n :

$$\sum_{x=3}^5 (2x + 3n) = 51$$

$$\rightarrow T_1 = 2(3) + 3n = 6 + 3n \therefore a = 3n + 6$$

$$T_2 = 2(4) + 3n = 8 + 3n$$

$$T_3 = 2(5) + 3n = 10 + 3n$$

$$\rightarrow T_2 - T_1 = 8 + 3n - (6 + 3n) = 2$$

$$T_3 - T_2 = 10 + 3n - (8 + 3n) = 2$$

$$\therefore d = 2 \therefore \text{AP}$$

$$\rightarrow \text{Number of terms} = (5 - 3) + 1 = 3$$

$$\rightarrow S_3 = \frac{3}{2} [2a + (n - 1)d] = 51$$

$$\therefore \frac{3}{2} [2(3n + 6) + (n - 1)(2)] = 51$$

$$\therefore \frac{3}{2} [6n + 12 + 2n - 2] = 51$$

$$\therefore \frac{2}{3} \left[\frac{3}{2} [6n + 12 + 2n - 2] \right] = \frac{2}{3} [51]$$

$$\therefore 8n + 10 = 34$$

$$\therefore 8n = 24$$

$$\therefore n = 3$$

SHORTCUT $\rightarrow$ first term ($x=3$); last term ($x=5$)

$$T_1 + T_2 + T_3 = 51$$

$$\therefore 6 + 3n + 8 + 3n + 10 + 3n = 51$$

$$\therefore 24 + 9n = 51$$

$$\therefore 9n = 27$$

$$\therefore n = 3$$

GP's

Eg. Find the value of n :

$$\sum_{r=1}^3 n(2^{r-1}) = 21$$

$$\rightarrow T_1 = n(2^{1-1}) = n(2^0) = n(1) \therefore a = n$$

$$T_2 = n(2^{2-1}) = n(2^1) = 2n$$

$$T_3 = n(2^{3-1}) = n(2^2) = 4n$$

$$\rightarrow \frac{T_3}{T_2} = \frac{4n}{2n} = 2$$

$$\frac{T_2}{T_1} = \frac{2n}{n} = 2 \therefore r = 2 \therefore \text{GP}$$

$$\rightarrow \text{Number of terms} = (3 - 1) + 1 = 3$$

$$\rightarrow S_3 = \frac{a(r^3 - 1)}{r - 1} = 21$$

$$\therefore \frac{a(r^3 - 1)}{r - 1} = 21$$

$$\therefore \frac{n(2^3 - 1)}{2 - 1} = 21$$

$$\therefore \frac{n(7)}{1} = 21$$

$$\therefore 7n = 21$$

$$\therefore n = 3$$

SHORTCUT $\rightarrow$ first term ($x=3$); last term ($x=5$)

$$T_1 + T_2 + T_3 = 21$$

$$\therefore n + 2n + 4n = 21$$

$$\therefore 7n = 21$$

$$\therefore n = 3$$

2.4 WRITING SERIES IN SIGMA NOTATION:

AP's

Eg. Write the series in sigma notation:

$$26 + 23 + 20 + 17 + 14 + 11 + 8 + 5$$

$$T_1 = 2(3) + 3n = 6 + 3n \therefore a = 3n + 6$$

$$T_2 = 2(4) + 3n = 8 + 3n$$

$$T_3 = 2(5) + 3n = 10 + 3n$$

$$\rightarrow T_2 - T_1 = 8 + 3n - (6 + 3n) = 2$$

$$T_3 - T_2 = 10 + 3n - (8 + 3n) = 2$$

$$\therefore d = 2 \therefore \text{AP}$$

$$\rightarrow \text{Number of terms} = (5 - 3) + 1 = 3$$

$$\rightarrow S_3 = \frac{3}{2} [2a + (n-1)d] = 51$$

$$\therefore \frac{3}{2} [2(3n+6) + (n-1)(2)] = 51$$

$$\therefore \frac{3}{2} [6n + 12 + 2n - 2] = 51$$

$$\therefore \frac{2}{3} \left[\frac{3}{2} [6n + 12 + 2n - 2] \right] = \frac{2}{3} [51]$$

$$\therefore 8n + 10 = 34$$

$$\therefore 8n = 24$$

$$\therefore n = 3$$

SHORTCUT $\rightarrow$ first term ($x=3$); last term ($x=5$)

$$T_1 + T_2 + T_3 = 51$$

$$\therefore 6 + 3n + 8 + 3n + 10 + 3n = 51$$

$$\therefore 24 + 9n = 51$$

$$\therefore 9n = 27$$

$$\therefore n = 3$$

$$\sum_{x=3}^5 (2x + 3n) = 51$$

GP's

Eg. Find the value of n :

$$\sum_{r=1}^3 n(2^{r-1}) = 21$$

$$\rightarrow T_1 = n(2^{1-1}) = n(2^0) = n(1) \therefore a = n$$

$$T_2 = n(2^{2-1}) = n(2^1) = 2n$$

$$T_3 = n(2^{3-1}) = n(2^2) = 4n$$

$$\rightarrow \frac{T_3}{T_2} = \frac{4n}{2n} = 2$$

$$\frac{T_2}{T_1} = \frac{2n}{n} = 2 \therefore r = 2 \therefore \text{GP}$$

$$\rightarrow \text{Number of terms} = (3 - 1) + 1 = 3$$

$$\rightarrow S_3 = \frac{a(r^3 - 1)}{r - 1} = 21$$

$$\therefore \frac{a(r^3 - 1)}{r - 1} = 21$$

$$\therefore \frac{n(2^3 - 1)}{2 - 1} = 21$$

$$\therefore \frac{n(7)}{1} = 21$$

$$\therefore 7n = 21$$

$$\therefore n = 3$$

SHORTCUT $\rightarrow$ first term ($x=3$); last term ($x=5$)

$$T_1 + T_2 + T_3 = 21$$

$$\therefore n + 2n + 4n = 21$$

$$\therefore 7n = 21$$

$$\therefore n = 3$$

4. BINOMIAL THEOREM

4.1 BINOMIAL THEOREM

$n!$ & $\binom{n}{r}$ NOTATION

$n!$ → n – factorial

$n!$ → $n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$

Eg. Evaluate $6!$

$$\begin{aligned} 6! &= 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 720 \end{aligned}$$

Calculator:



Eg. Evaluate $\frac{12!}{10!}$

$$\frac{12!}{10!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= \frac{12 \times 11 \times \cancel{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}}{\cancel{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}}$$

$$= 720$$

Eg. Evaluate $\frac{13!}{11!} - \frac{6!}{4!}$

$\binom{n}{r}$ notation $\rightarrow$ (read as "***n choose r***") which represents a ***binomial coefficient***. It calculates the number of ways to choose r items from a set of n distinct items without considering the order, and it provides the coefficients used to expand binomial expressions like $(a + b)^n$

$$\# \binom{n}{r} = \frac{n!}{r! \times (n-r)!}$$

NB!
DO NOT WRITE AS $\binom{n}{r}$ A FRACTION.
THERE IS NO LINE BETWEEN n and r .
IT MUST LOOK LIKE A "COLUMN VECTOR"

Eg. Evaluate $\binom{6}{4}$

$$\begin{aligned} \binom{6}{4} &= \frac{6!}{4! \times (6-4)!} = \frac{6!}{4! \times 2!} \\ &= \frac{6 \times 5 \times \cancel{4 \times 3 \times 2 \times 1}}{\cancel{4 \times 3 \times 2 \times 1} \times (2 \times 1)} = \frac{6 \times 5}{2} = \frac{30}{2} = 15 \end{aligned}$$

PASCAL'S TRIANGLE

The Pascal Triangle can be used as an alternative to $\binom{n}{r}$ notation in order to find the coefficients of the expansion.

ROW 1 $\rightarrow n = 0$	$\rightarrow$	1
ROW 2 $\rightarrow n = 1$	$\rightarrow$	1 1
ROW 3 $\rightarrow n = 2$	$\rightarrow$	1 2 1
ROW 4 $\rightarrow n = 3$	$\rightarrow$	1 3 3 1
ROW 5 $\rightarrow n = 4$	$\rightarrow$	1 4 6 4 1
ROW 6 $\rightarrow n = 5$	$\rightarrow$	1 5 10 10 5 1
ROW 7 $\rightarrow n = 6$	$\rightarrow$	1 6 15 20 15 6 1
ROW 8 $\rightarrow n = 7$	$\rightarrow$	1 7 21 35 35 21 7 1
ROW 9 $\rightarrow n = 8$	$\rightarrow$	1 8 28 56 70 56 28 8 1

Eg. Evaluate $\binom{6}{4}$

1						
1	1					
1	2	1				
1	3	3	1			
1	4	6	4	1		
1	5	10	10	5	1	
1	6	15	20	15	6	1
0	1	2	3	4	5	6

r always start at 0 and end when r is the same value as n .

$$\binom{6}{0} \binom{6}{1} \binom{6}{2} \binom{6}{3} \binom{6}{4} \binom{6}{5} \binom{6}{6}$$

Take note that the answer is the same as the one calculated above $\rightarrow 15$

$$\therefore \binom{6}{4} = 15$$

4.2 BINOMIAL EXPANSION

Pattern to expand $(a + b)^n$ using binomial Pascal's Triangle.

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \binom{n}{3} a^{n-3}b^3 + \dots \binom{n}{r} a^{n-r}b^r + \binom{n}{n} b^n$$

This is given in the formula sheet

Easy way to do binomial expansion:

Eg. Expand $(a + b)^4$

Start with the **a** term → Power 4 means there will be 5 terms in expansion, but we write 4 **a's**

$$a^4 \quad a^3 \quad a^2 \quad a^1$$

Now from the right side, start doing the same for the **b's** starting next to the last **a** going backwards.

$$a^4 \quad a^3b^1 \quad a^2b^2 \quad ab^3 \quad b^4$$

Next, we fill in the coefficients. The easiest method to use is Pascal's Triangle.

				1					0
			1		1				1
		1		2		1			2
	1		3		3		1		3
1		4		6		4		1	4

$$1a^4 \quad 4a^3b \quad 6a^2b^2 \quad 4ab^3 \quad 1b^4$$

Lastly, we fill in the missing operators between the terms.

$$a^4 \quad + \quad 4a^3b \quad + \quad 6a^2b^2 \quad + \quad 4ab^3 \quad + \quad b^4$$

NOTE: Looking at the question $(a + b)^4$

If there is a + between the 2 terms, all terms are + in the expansion.

If there is a - between the 2 terms, the operators will look like: + - + - + -

If the question was $(a - b)^4$,

$$+ \quad a^4 \quad - \quad 4a^3b \quad + \quad 6a^2b^2 \quad - \quad 4ab^3 \quad + \quad b^4$$

$$= a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4$$

Eg. Expand $(x + y)^7$ using binomial Pascal's Triangle.

→ Use the row of $n = 7$: 1 7 21 35 35 21 7 1

$$x^7 + 7x^6y + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + y^7$$

Eg. Expand $\left(3x - \frac{y}{2}\right)^4$ using the binomial expansion.

→ Use the row of $n = 4$: 1 4 6 4 1

$$\begin{aligned} & (3x)^4 \left(\frac{1}{2}y\right)^0 - 4(3x)^3 \left(\frac{1}{2}y\right)^1 + 6(3x)^2 \left(\frac{1}{2}y\right)^2 - 4(3x)^1 \left(\frac{1}{2}y\right)^3 + (3x)^0 \left(\frac{1}{2}y\right)^4 \\ & 81x^4 - 108x^3 \left(\frac{1}{2}y\right) + 54x^2 \left(\frac{1}{4}y^2\right) - 12x \left(\frac{1}{8}y^3\right) + \left(\frac{1}{16}y^4\right) \\ & 81x^4 - 54x^3y + 13\frac{1}{2}x^2y^2 - 1\frac{1}{2}xy^3 + \frac{1}{16}y^4 \end{aligned}$$

Eg. (a) Find, in ascending powers of a , the first 3 terms in the expansion of $\left(2 - \frac{a}{4}\right)^{10}$.

(b) Hence, obtain the coefficient of a^2 in the expansion of $(1 + 2a - 3a^2) \left(2 - \frac{a}{4}\right)^{10}$

Ex. In the expansion of $(3 - 2x)\left(1 + \frac{x}{2}\right)^n$, the coefficient of x is 7.

Find the value of the constant n and hence find the coefficient of x^2 .

[illegible]